

The evaluation of overall stability of retained fills, as well as earth slopes with or without a shallow or deep foundation unit, should be investigated at the strength and extreme event limit states based on the Strength I and Extreme Event I load combinations using γ_p for overall stability as specified in Table 3.4.1-2 and an appropriate resistance factor as specified in Article 11.6.3.7. If the externally applied load(s) contribute to the stability of the slope, the externally applied load should not be included in the load combination used for design of the slope. However, if the foundation element is designed to resist the additional load induced on the element by the slope instability, design of the slope for overall stability may include the resisting load applied to the slope by the foundation element.

The evaluation of foundation movements as defined in Article 10.5.2.2 shall be conducted in accordance with Article 10.6.2.4 using the load factor, γ_{SE} , specified in Table 3.4.1-5.

For structural plate box structures complying with the provisions of Article 12.9, the live load factor for the vehicular live loads, *LL* and *IM*, shall be taken as 2.0.

low load factor as appropriate. The algebraic sums of these products are the total force effects for which the bridge and its components should be designed.

PS, *CR*, *SH*, *TU*, and *TG* are superimposed deformations as defined in Article 3.12. For prestressed members in typical bridge types, secondary prestressing, creep, and shrinkage are generally designed for in the service limit state. In concrete segmental structures, *CR* and *SH* are factored by γ_p for *DC* because analysis for time-dependent effects in segmental bridges is nonlinear. Abutments, piers, columns, and bent caps are to be considered as substructure components.

Because force effects due to uniform temperature change, temperature gradient, prestressing, creep, and shrinkage are the result of imposed deformation, member and foundation stiffness can have a significant impact on calculated values. Pursuant to Articles 4.5.1 and 4.5.2, and for certain limit states, consideration may be given to the effects of cracking on the stiffness of concrete members placed in flexure as well as the flexibility of foundation systems.

The calculation of displacements for *TU* utilizes a factor greater than 1.0 to avoid undersizing joints, expansion devices, and bearings.

Loads on foundations due to forces caused by slope instability should be determined in accordance with Liang (2010), or Vessely et al. (2007) and Yamasaki et al. (2013). Illustrations of this are provided in Figures C3.4.1-1 and C3.4.1-2.

Available slope stability design procedures and programs produce a single factor of safety, *FS*, using limit equilibrium. However, because the specific location of the critical failure surface varies depending on the overall slope geometry, subsurface stratigraphy, and external loads acting on the failure mass, and since a portion of the gravity force acting on the failure mass contributes to load and a portion contributes to resistance, there is no effective way to apply a load factor directly to the loads within the soil failure mass. Hence, the load factor of 1.0 for overall stability in Table 3.4.1-2 is combined with a resistance factor that is equal to $1/FS$ as provided in Article 10.5.5.2.1 and Article 11.6.3.7. Furthermore, slope stability analysis uses limit equilibrium concepts rather than slope deformation to assess shear failure of the soil. Therefore, overall stability is conducted for strength and extreme event limit states.

If reinforcement elements are placed through the slope failure mass to improve its stability, they should be considered to be resisting elements. Appropriate resistance factors should be applied to the resistance derived from those elements. For foundations, resistance factors that should be used are as specified in Article 10.5.5.2. For soil reinforcement elements such as are normally associated with walls, resistance factors that should be used are as specified in Article 11.5.7.